Operational validity is not a spacetime embedding

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Abstract

Finite process functions are classical rules for which every tuple of local functions has exactly one history. For a stated canonical extension, that condition is necessary and sufficient for the closed loop to stay unitary under every local unitary and every matched finite ancilla. Dilation then yields an operationally valid process under local instruments. These are theorems about maps on finite-dimensional spaces tied to alphabets. They do not embed the loop in a relativistic spacetime, produce a local Hamiltonian, or answer the distinct question of when a cyclic information-theoretic structure can sit inside an acyclic spacetime. A known quantum board, read in one named basis, also shows that operational purity need not return the classical record to a mixture of process functions. The paper states what the operational results establish and where the spacetime question remains untouched.

Keywords. process functions; indefinite causal order; operational validity; spacetime embedding

1 Two questions that share a vocabulary

The process-matrix framework assigns probabilities to local experiments without assuming that the laboratories sit in one fixed causal order (Oreshkov et al., 2012). Its deterministic classical points are process functions: rules from the parties' outputs to their inputs such that every choice of local functions has exactly one consistent history (Baumeler and Wolf, 2016). Baumeler et al. (2019) show that deterministic dynamics of this kind, compatible with arbitrary local operations, admit a reversible realization inside that classical theory, including arrangements in which three parties are each in the past and the future of the others. A later result, reported in an unpublished technical note and set out again below, says something more specific. For finite alphabets, a rule is a process function if and only if one canonical reversible extension remains unitary when every local unitary, on every matched finite ancilla, is inserted. From that unitarity, ordinary instruments follow by dilation and partial trace.

It is natural to hear these sentences as spacetime physics. "No causal order," "each in the past of the others," and "unitary closed loop" are phrases that, in general relativity, belong to curves, regions, and Hamiltonians. The hearing is a mistake about the subject of the theorems. The theorems concern alphabets, functions, and operators. A second literature asks when a cyclic information-theoretic structure can be embedded in an acyclic relativistic spacetime, and proves no-go results under stated assumptions (Vilasini and Renner, 2024). That is a different question. This paper keeps the two questions apart. It says what universal intervention-consistency and the canonical unitary composition establish, and why operational validity does not establish a spacetime implementation.

The argument is not a survey of indefinite causal order, and it is not a theory of metaphysical recombination of events. One running distinction is enough. An intervention class is a set of maps the parties may insert. A history is a solution of the resulting equations. A spacetime implementation would be a further story: regions, a metric or a causal structure of spacetime, and a dynamics that realizes

the operators as the physics of those regions. The first two items are settled, for finite alphabets, by the equivalence below. The third is not mentioned by the equivalence.

Related drafts of the technical note are already public. The journal's own guidance treats preprint deposit as compatible with review and as something that can defeat anonymity. This manuscript therefore does not claim that anonymity is secure. The proofs needed for the argument are written out. No repository address is printed. Self-identifying citation of the note is omitted in the review text where a citation would only name the author; the declarations sheet records the provenance.

2 What the operational equivalence says

Let each party k have finite nonempty input and output alphabets I_k and O_k. A local intervention, in the deterministic sense, is a function $f_k : I_k \to O_k$. A rule w sends a tuple of outputs to a tuple of inputs. The history equation says that the outputs, passed through w and then through the local functions, return the same outputs. The rule is a process function when every tuple of local functions has exactly one solution (Baumeler and Wolf, 2016). The quantifier ranges over local functions. It does not range over spacetime regions, and it does not say that the parties are free in any metaphysical sense. It says which global rules are admitted if every local function of the given type is to be installable.

The three-bit Lugano rule

$$w(a, b, c) = (\neg b \wedge c,\ \neg c \wedge a,\ \neg a \wedge b)$$

is a standard example (Baumeler and Wolf, 2016; Baumeler et al., 2019). Each input depends on the other two outputs. There is no order of the three parties in which each depends only on earlier parties. For the intervention in which every party copies its input, the only solution is $(0, 0, 0)$: the case $a = 1$ forces c to be both 1 and 0. One intervention does not prove the universal claim. It shows what a history is. The rule is a process function by the finite definition, as recorded in the technical note.

Add a source, whose basis is indexed by input tuples, and a sink, whose basis is indexed by output tuples. For every rule w, not only for process functions,

$$U|o\rangle|e\rangle = |w(o) + e\rangle|o\rangle$$

is a permutation of a finite basis, once each input alphabet is read as a cyclic group of the same order. A finite permutation is unitary. Bare unitarity of U is therefore not the process-function condition, and it is not a physical Hamiltonian. It is bookkeeping that exists for inconsistent rules as well.

The quantum test inserts local unitaries. Let V_k map $I_k \otimes X_k$ unitarily onto $O_k \otimes Y_k$, with finite ancillas and $|I_k|\,|X_k| = |O_k|\,|Y_k|$. The closed loop M is the operator obtained by wiring each V_k into U. Its matrix elements factor over parties. The unpublished note, and the argument in the next paragraph, show that M is unitary for every such choice if and only if w is a process function.

If w is a process function and two output strings differ, some party has different outputs and the same input on those two strings. Otherwise a single local intervention would have both strings as histories. The Gram matrix of M is a product over parties. Where the two strings give that party the same input, a change of variables turns the factor into an inner product of rows of V_k. Where exclusivity supplies equal inputs and unequal outputs, the factor is the inner product of two different rows and vanishes. Thus $MM^\dagger$ is the identity. The dimension match makes M square, so M is unitary. Conversely, if some intervention has $N \neq 1$ histories, the maps $V_k|i, x\rangle = |f_k(i) + x,\ i\rangle$ are unitary, and one column of M has squared norm N. The equivalence uses both directions. Orthogonality is not a column-norm argument, and the column norm is not a substitute for exclusivity.

The ancillas may enter the state already entangled with each other and with the source. That is a fact about the state, not a further assumption on w. The equivalence is for this canonical U and this class of

tests. It is not the claim that every unitary dilation whatsoever exists if and only if w is a process function. A local unitary that fails to be diagonal in the computational basis mixes basis states the classical table treated as distinct. The theorem says that a process function survives every such mixing, and that a rule which survives every such mixing is a process function. Survival is not automatic from the permutation U.

3 Validity is still an operational notion

Unitarity of M is a condition on maps that are themselves unitary. Laboratories also perform instruments that are not unitary. The passage is the usual finite-dimensional one. A completely positive trace-preserving map dilates to an isometry into a larger output space; with finite junk spaces one extends the isometry to a unitary; the lifting theorem supplies a unitary on the enlarged loop; the environments are prepared in a fixed state and discarded. Partial trace preserves complete positivity and the trace. The link product is multilinear, so the identity extends from the dilated unitaries to the original instruments. An instrument is a family of such maps.

Call the result operational validity: for every tuple of local instruments of this finite-dimensional kind, the induced map from the source side to the sink side is a valid quantum process in the process-matrix sense (Oreshkov et al., 2012; Araújo et al., 2017). The word "valid" here means normalization and complete positivity of that higher-order map. It does not mean "occurs in a spacetime." The dilation does not choose a metric. Discarding an environment is not a claim about a horizon or a wormhole. The finite-dimensional padding is essential to the form of the argument written here. The technical note treats further infinite-dimensional bookkeeping separately; those extensions are not used.

Two intervention classes have now been named, and they must not be collapsed. Deterministic functions are the process-function quantifier. Local unitaries on matched ancillas are the canonical equivalence. Instruments are the image of the unitary case under dilation and partial trace. A fourth class, stochastic classical maps, asks only that a classical conditional distribution assign total probability one under every product of local stochastic maps. A process function induces such a distribution, concentrated on its unique history. The converse is false, and is not needed until Section 5. Each class is a set of maps on alphabets or on finite-dimensional Hilbert spaces associated with alphabets. None is a set of spacetime regions.

Baumeler et al. (2019) already separate a classical reversible realization from an assumption that some local operations must be forbidden. Their result is inside the classical process theory. The canonical equivalence adds the quantum test for one explicit extension, and the dilation adds instruments. Someone who says "the laboratories may perform any local quantum operation, and the global map stays physical" is, if they mean this theorem, saying something true and operational. Someone who adds "and therefore there is a region of spacetime in which those laboratories sit on a closed causal curve" has changed the subject.

4 The spacetime question is a different subject

A spacetime implementation, as the phrase is used here, would supply at least the following. There are regions, or events, in a relativistic spacetime. There is a causal structure, or a metric, relative to which some pairs of regions are spacelike and some are causally ordered. The operational maps are the reduced description of a dynamics on that spacetime, or of quantum fields, or of a local Hamiltonian interaction, after the regions are specified. None of these data appear in U, in M, or in the dilation. The alphabets do not name points. The source and the sink are extra factors in a tensor product, not asymptotically early and late surfaces picked out by a Killing time.

Vilasini and Renner (2024) study the embedding question directly. Their title states no-go results for embedding cyclic information-theoretic structures in acyclic spacetimes. This paper does not re-prove those theorems and does not claim a stronger impossibility than they do. It claims a narrower negative. The canonical lift is not a counterexample to an embedding no-go, because the canonical lift is not an embedding. An operator identity on a finite-dimensional space does not assert that the space is the Hilbert space of a region in Minkowski spacetime, or in a spacetime with a closed timelike curve. If a later construction embeds this particular U in a spacetime, that construction will have to add the geometric data and discharge whatever hypotheses the no-go theorems use. The operational equivalence will remain what it is, whether or not that geometric construction exists.

The process-matrix phrase "no causal order" is itself operational (Oreshkov et al., 2012). It means that the probabilities are not those of a process that is a mixture of processes with a fixed order of the laboratories, or, in the deterministic case, that the functional dependence among the parties is not acyclic. Baumeler et al. (2019) use "past" and "future" for the direction of that functional dependence: a party's input depends on another's output. That is a statement about the rule w. It becomes a statement about relativistic past and future only with a further identification of parties with regions and of dependence with causal connectibility. The identification is not a theorem of the process function. It is an interpretive hypothesis, and it is the hypothesis the embedding literature has to assess rather than assume.

There is a familiar way to imitate the event list without the geometry. Post-select a circuit on the histories that satisfy the fixed-point equation. Conditional on success, the samples can match the list of outputs. Post-selection is a filter on runs of an ordinary circuit. It is not the unitary M, which is deterministic on the source-to-sink space when the lift applies, and it is not a spacetime. Using post-selection as a picture of the theorem confuses a sampling trick with the dilation.

Nothing here denies that some processes in the process-matrix framework might someday be given spacetime models, or that some laboratory experiments with controlled optical paths realize specific processes such as the quantum switch. Those are existence claims about particular constructions. They are not read off the biconditional "process function if and only if the canonical loop is unitary for all local unitaries." A biconditional about alphabets does not become a construction in geometry by being true.

5 What the classical record of a pure process is not

Operational purity, in the sense of Araújo et al. (2017), is unitarity of the closed loop for every local unitary. The canonical equivalence says that process functions meet that standard for one extension, and that only they do. Mixtures of process functions meet it as well, by a control register that selects the component and is routed to the sink: the loop is block diagonal and hence unitary. Discarding the sink and reading a fixed basis leaves the classical mixture. So purity is compatible with ignorance over deterministic consistent rules.

The converse reading is the one to refuse. It says that if a process is pure, then the classical distribution it induces is a mixture of process functions. That reading is false for a single specified case, and the case should stay small so that it is not asked to carry a theory of spacetime. The technical note gives an explicit three-party binary process, induced in the computational basis by a known quantum board and a stated source state, whose distribution p satisfies

$$L(p) = \frac{1}{4} - \frac{3}{16}\sqrt{2} < 0,$$

while $L \geq 0$ on every three-bit process function and hence on their convex hull. The board is identified in that note, up to a local frame, with a time-reversed variant of a known Lugano purification. It is not

presented here as a newly discovered quantum process. The separation is basis-relative. In another basis, a pure process of the same family can induce a mixture. What is excluded is one sentence about one record: that this computational-basis distribution is ignorance of a process function. What is not excluded is classical physics, a spacetime model of the board, or a hidden variable of some other kind.

The witness matters for the present argument because purity is the operational property that most tempts a geometric reading. If purity meant "there is a reversible spacetime process underneath, whose classical shadow is a mixture of consistent deterministic histories," the negative value of L would be an embarrassment for that slogan. The slogan was never the theorem. The theorem is unitarity of M. The value of L shows that unitarity of M, plus a computational-basis reading, does not restore the deterministic classical hull. Spacetime structure is not restored either, because it was not derived.

A full characterization of which classical processes arise from pure processes is open. The one inclusion used here is that the convex hull of the process functions is a proper subset of the processes induced in the stated way. Certificates that some valid classical processes fail to be purifiable are discussed in the technical note and are not re-proved. They are not needed. The spacetime gap does not depend on how large the gap between purity and the classical hull is. It depends on the absence of geometric data in the operational theorems.

6 Objections

The first objection says that the parties are laboratories, and laboratories are regions, so the theorem is already about spacetime. A laboratory, in the process formalism, is a slot at which an instrument is chosen (Oreshkov et al., 2012). The slot has input and output Hilbert spaces. It does not come with coordinates. One may intend, when applying the formalism, that the slots correspond to benches in a room. The intention is an extra model. The equivalence does not mention benches. If the objection means only that the formalism was designed with laboratories in mind, that is a remark about motivation, not a derivation of a metric.

The second objection says that unitarity is time-reversal symmetry, so the lift is a result about temporal symmetry of the world. Unitarity of M is invertibility of a linear map between finite-dimensional spaces, with $M^\dagger = M^{-1}$. The inverse of U exchanges the bookkeeping roles of source and sink. That is a symmetry of an operator. Temporal symmetry of a world would say that the dynamical laws do not distinguish a time orientation, or that a process and its time reverse are both allowed by the same spacetime theory. The operator symmetry can be proposed as a formal analogue. The proposal is an interpretation. It is not the content of $MM^\dagger = 1$. In particular, it does not say which spacetime time-reversal, acting on which fields, U represents.

The third objection says that operational validity is the physicist's notion of a physical process, so a demand for spacetime is a demand for metaphysics beyond physics. The objection uses "physical" for two standards. The process-matrix standard is that local instruments compose to a trace-preserving completely positive map. That standard is met, for process functions, by the dilation argument. The relativistic standard is that the same composition arise from local dynamics in a spacetime with a stated causal structure. Experiments and theories can meet the first standard without a completed answer to the second. Optical implementations of particular processes are existence proofs for those processes under those laboratory identifications. They are not a general translation of every process function into a solution of Einstein's equation. Refusing the general translation is not a retreat from physics to metaphysics. It is a refusal to treat one formalism's notion of validity as every other formalism's notion of realization.

The fourth objection says that if three parties are each in the past and the future of the others, as Baumeler et al. (2019) put it, then a closed causal curve has already been demonstrated. Their

abstract states the result for deterministic dynamics on degrees of freedom associated with regions, inside the process theory, with "past" and "future" tracking that dependence. The demonstration that such dependence is compatible with arbitrary local operations is a result about consistency of the process theory. It is not, by itself, a solution of the Einstein equation with a closed timelike curve, and it is not a claim that the degrees of freedom are fields on such a curve. The present equivalence does not strengthen that geometric reading. It only characterizes which classical rules make one quantum extension unitary.

The fifth objection says that the basis-relative witness is too weak to matter: change the frame and the classical mixture may return, so purity and process functions are the same physics. They are the same quantum board only in the sense the note already grants, that the board is a known one up to a frame. The classical record is not frame-independent. A laboratory that measures in the computational basis and records p does not record a mixture of process functions, even if a different choice of local frames would have produced a mixture. Operational validity of the quantum process does not tell the laboratory which classical record it will see. The frame is part of the operational description. Leaving it out is how purity gets described as if it were a spacetime fact rather than a fact about operators and a measurement.

7 Conclusion

Universal intervention-consistency, for finite alphabets, is the condition that every tuple of local functions has exactly one history. For the canonical source-sink extension, that condition is necessary and sufficient for unitarity under every local unitary and every matched finite ancilla. Dilation and partial trace then give operational validity under local instruments. These results fix the relationship between a classical admissibility condition and a quantum composition condition inside the process formalism. They do not supply regions, a metric, a local Hamiltonian, or an embedding in an acyclic spacetime. The embedding question remains the question studied by Vilasini and Renner (2024), not a corollary of $MM^\dagger = 1$. A known board, read in the computational basis, shows in addition that operational purity does not mean the induced classical process is a mixture of process functions. That limit is basis-relative and does not close the spacetime gap. It prevents a shorter misreading, on which the unitary loop was already the reversible spacetime process whose classical shadow had to be a consistent deterministic mixture.

Competing interests

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Preparation of this draft, including organization of the argument and wording of the proofs, was assisted by Grok Bot, a Cursor cloud agent. The assistance included generative composition, not only copy editing. No version identifier for the tool is recorded here, and none is invented. The assistant is not an author. The journal's own AI-policy page was not retrieved in full when this draft was prepared; a publisher-level statement requires disclosure of AI writing tools in the methods or a similar section. This section is that disclosure. Whether the journal later requires a different slot is unresolved. Author verification is pending.

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